

Black and Scholes: The Trillion Dollar Equation

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1 Introduction

This paper explores the detailed derivation and solution to the Black and Scholes option pricing problem [1]. Chapters 2 through 4 focus on a brief introduction to mathematical prerequisites in stochastic calculus essential in our derivation and solution methods. Chapter 5 is dedicated to deriving the Black-Scholes PDE using a riskless replicating portfolio and solving the Black-Scholes PDE using alternative methods, these being the heat equation transformation and Feynman–Kac solutions, that are equivalent and both satisfy the Black and Scholes PDE. Chapter 6 briefly covers realised and implied volatility, how to calculate them, and their roles in option pricing. Chapter 7 explores comparative statistics (Delta, Gamma, and Theta) of the Black-Scholes PDE for European plain vanilla call and put options; we also explore option pricing trading strategies that can be used in idealistic scenarios to make risky profit. Chapter 8 outlines the assumptions that underpin the model, and Chapter 9 provides a detailed conclusion on all the above.

2 Brownian Motion

A continuous stochastic process, $B(t); \{t \geq 1\}$ is a Brownian motion if and only if

- **Zero Initial Value.** $B(0) = 0$
- **Standard Normally Distributed.** $B(t) \sim N(0, t)$, that is, realisations of the process are normally distributed with first moment (mean) $\mu = 0$ and centred second moment (variance) $\sigma^2 = t$. So we have the Gaussian Brownian density function $f_{B(t)}(x) = \mathcal{N}(B(t)|\mu = 0; \sigma^2 = t)$.
- **Gaussian increments.** That is, denoting increments using i, j for any $t_i > t_j, (i, j) \in \mathbb{N}$ we have density $f_{B(t_i)}(x) - f_{B(t_j)}(x) = \mathcal{N}(B(t_i)|\mu = 0; \sigma^2 = t_i) - \mathcal{N}(B(t_j)|\mu = 0; \sigma^2 = t_j) = \mathcal{N}(B(t)|\mu = 0; \sigma^2 = t_i - t_j)$. More simply $B_i - B_j \sim N(0, t_i - t_j)$
- **Independent increments.** So $f_{B(t_i)|B(t_j)}(x) = f_{B(t_i)}$.

3 Ito's Process

An Ito process is a function $f(X_t, t)$ that is twice differentiable in price, X_t , and once in time, t . An Ito process's evolution is governed by an SDE, which is generally written in equation 1.

$$dX_t = a(X_t, t)dt + b(X_t, t)dB(t); B(t) \sim N(0, 1) \quad (1)$$

We can integrate our Ito process, $f(X_t, t)$, according to the SDE, 1, to get the solution of the SDE.

$$X_t = X_0 + \int_0^t a(X_t, t)dt + \int_0^t b(X_t, t)dB(t) \quad (2)$$

We are concerned with the study of how Ito processes evolve. However, we cannot use standard calculus to evaluate the rate of change of an Ito process due to the stochastic term in the SDE, $\sigma dB(t)$. Proceeding in the standard case we get.

$$\begin{aligned} df &= \frac{\partial}{\partial x} f(X_t, t)dx + \frac{\partial}{\partial t} f(X_t, t)dt + \frac{\partial}{\partial B} f(X_t, t)dB(t) \\ &= f_x (a(X_t, t)dt + b(X_t, t)dB(t)) + f_t dt \\ &= (a(X_t, t)f_x + f_t) dt + b(X_t, t)dB(t) \end{aligned} \quad (3)$$

To attempt to measure the evolution of our Ito process, $f(X_t, t)$, we apply the chain rule, which is a logical choice given we have a function of two variables, X_t and t . As we will see the standard calculus definition in equation 3 is incorrect and does not account for the quadratic variation term.

4 Ito's Lemma

Ito's lemma is the stochastic version of the chain rule which we can use to correctly define the evolution of any valid ito-process.

4.1 Ito's Lemma From First Principles

We proceed with the derivation, as follows.

$f(X_t, t)$ as before is assumed to be a valid ito-process, twice differentiable in X_t and once in t . So our function is multivariate and is a mapping $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. We can use a multivariate Taylor expansion in the inputs to approximate a small perturbation of the function (we are effectively attempting to approximate the derivative of the Ito-process). For two variables of one function the Taylor series approximation is

$$f(x, y) = \sum_{k=0}^{n=\infty} \frac{1}{k!} \left(\frac{\partial}{\partial x}(x-a) + \frac{\partial}{\partial y}(y-b) \right)^{(n)} f(x, y) \Big|_{x=a, y=b} \quad (4)$$

We perturb both inputs.

$$f(X_t, t) \rightarrow f(X_t + \Delta x, t + \Delta t) \quad (5)$$

Where $\Delta x = X_{t+\Delta t} - X_t$. We will evaluate 5 about the original points (X_t, t) . The Taylor expansion is as follows.

$$\begin{aligned} f(X_t + \Delta x, t + \Delta t) &= \sum_{k=0}^{n=\infty} \frac{1}{k!} \left(\frac{\partial}{\partial x} \Delta x + \frac{\partial}{\partial y} \Delta t \right)^{(n)} f(X_t + \Delta x, t + \Delta t) \Big|_{x=X_t, y=t} \\ &= \frac{1}{0!} f + \frac{1}{1!} (f_x \Delta x + f_t \Delta t) + \frac{1}{2!} (f_{xx} (\Delta x)^2 - 2f_{xt} \Delta x + \Delta t f_{tt} (\Delta t)^2) + \dots \end{aligned} \quad (6)$$

In standard calculus, higher-order terms vanish, although not here. Using 1 we have

$$(\Delta x)^2 = a^2(X_t, t)(\Delta t)^2 + 2a(X_t, t)b(X_t, t)\Delta t \Delta B_t + b^2(X_t, t)(\Delta B_t)^2$$

Where

$$(\Delta t)^2 = 0 \quad (7)$$

$$\Delta t \Delta B(t) = 0 \quad (8)$$

$$(\Delta B(t))^2 = (\sqrt{\Delta t})^2 = \Delta t \quad (9)$$

We substitute identities 7, 8 and 9 into the expansion of $(\Delta x)^2$ and the Taylor expansion in 6. If we then evaluate the limit of this process, so $\Delta \rightarrow 0 \implies \Delta \approx dx$ and let $f(X_{t+\Delta t}, t + \Delta t) - f(X_t, t) = df$ we arrive at Ito's lemma for an Ito-process.

$$\begin{aligned}
df &= (f_x dx + f_t dt) + \frac{1}{2} f_{xx} (dx)^2 \\
&= f_x(a(X_t, t)dx + b(X_t, t)dB(t)) + f_t dt + \frac{1}{2} f_{xx} b^2(X_t, t)dt \\
&= \left(a(X_t, t)f_x + f_t + \frac{1}{2} b^2(X_t, t)f_{xx} \right) dt + b(X_t, t)f_x dB(t)
\end{aligned} \tag{10}$$

4.2 Arithmetic Brownian Motion (ABM)

This is an Ito process that evolves according to an SDE of the following form

$$dX_t = \mu dt + \sigma dB(t); B(t) \sim N(0, t) \tag{11}$$

With solution

$$X_t = X_0 + \mu t + \sigma B(t) \tag{12}$$

To find the distribution of the Arithmetic Brownian Motion (ABM), we can use a change of variables. Let the solution of the ABM in 11 be Y_t such that $Y_t = g(B(t)) \iff g^{-1}(Y_t) = B(t)$. Then to get the new distribution form we must solve.

$$f_{Y_t}(y)dy = f_{B(t)}(x)dx \tag{13}$$

We can re-write 13 in an equivalent form, to get

$$f_{Y_t}(y) = f_{B(t)}(x = g^{-1}(B(t))) \left| \frac{d}{dy} g^{-1}(B(t)) \right| \tag{14}$$

Now we plug in and simplify to uncover the transformed distribution. Equation 15 defines the inverse transformation,

$$g^{-1}(Y_t) = \frac{y - X_0 - \mu t}{\sigma}; \frac{dB(t)}{dy} = \frac{1}{\sigma} \tag{15}$$

Substituting in $B(t) \sim N(0, t)$ we get

$$f_{Y_t}(y) = \frac{1}{\sigma t \sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \left(\frac{y - (X_0 + \mu t)}{\sigma t} \right)^2 \right\} \tag{16}$$

It is clear from the density, equation 16, that $f_{Y_t}(y)$ is also a normal distribution, specifically.

$$X_t \sim N(X_0 + \mu t, \sigma^2 t) \tag{17}$$

Equation 17 is the distribution of the solution to the ABM defined by 11. We can see that the ABM is linear in time, t . If we imagine X_t as price, then by properties of the normal, price can take on negative values, which we know to be impossible in real markets. Therefore, we can conclude the ABM is a poor candidate process for modelling asset prices.

4.3 Geometric Brownian Motion (GBM)

A Geometric Brownian Motion is a SDE of the form

$$dS_t = \mu S_t dt + \sigma S_t dB(t); B(t) \sim N(0, 1) \quad (18)$$

Where $X(t) = \log(S_t)$ is a function of S_t , so X_t here is an ito process $f(S_t, t)$. We can use Ito's lemma 10 to find the evolution of the process, and integrate to find the solution that satisfies the SDE. We apply Ito's lemma to the function $X_t = f(S_t, t) = \log(S_t)$.

$$f_t = 0, f_x = \frac{1}{S_t}, f_{xx} = -\frac{1}{S_t^2} \quad (19)$$

Substituting into 10 we get

$$\begin{aligned} df(S_t, t) &= \left(\mu S_t \frac{1}{S_t} - \frac{1}{2} \sigma^2 S_t^2 f_{xx} \right) dt + b S_t \frac{1}{S_t} dB(t) \\ &= X_0 + \mu t - \frac{1}{2} \sigma^2 t + \sigma B(t) \\ X_t &= X_0 + \mu \int_0^t dt - \frac{1}{2} \sigma^2 \int_0^t dt + \sigma \int_0^t dB(t) \\ &= X_0 + \mu t - \frac{1}{2} \sigma^2 t + \sigma B(t) \end{aligned} \quad (20)$$

Equation 20 is the solution to the Ito process $f(S_t, t)$. Equation 20 is effectively an affine transformation of a Gaussian-distributed random variable, so we can use linearity of expectation (or using change of variables via the probability density transform) and properties of the Brownian motion; we know that X_t is normally distributed.

$$X_t \sim N \left(X_0 + \left(\mu - \frac{1}{2} \sigma^2 \right) t, \sigma^2 t \right) \quad (21)$$

We now transform back to get the solution in S_t space.

$$\begin{aligned} S_t &= \exp \left\{ X_0 + \mu t - \frac{1}{2} \sigma^2 t + \sigma B(t) \right\} \\ &= S_0 \exp \left\{ \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B(t) \right\} \end{aligned} \quad (22)$$

Equation 22 is the famous solution to the GBM Ito process. To identify the distribution of the solution to the Ito process, $X(t) = \log(S_t)$, that follows a GBM, we again use the probability density transform.

$$g(X_t) = e^{X_t} = S_t \iff g^{-1}(S_t) = \log(S_t) = X_t$$

Substituting in probability density transform we have

$$\begin{aligned} f_{S_t}(S_t) &= f_{X_t}(\log(S_t)) \left| \frac{d}{dS_t} \log(S_t) \right| \\ &= f_{X_t}(\log(S_t)) \frac{1}{S_t}, X_t \text{ follows distribution ??} \\ &= \frac{1}{S_t \sqrt{2\pi\sigma^2 t}} \exp \left\{ -\frac{1}{2\sigma^2 t} (\log(S_t) - \log(S_0) - (\mu - \frac{1}{2}\sigma^2 t))^2 \right\} \end{aligned} \quad (23)$$

We know that the exponential of a Gaussian-distributed random variable, X_t , is log-normally distributed; the stock price, S_t follows a log normal distribution with density given by equation ???. We denote the distribution of S_t as

$$S_t \sim \text{Lognormal}(\log(S_0) + (\mu - \frac{1}{2}\sigma^2)t, \sigma^2 t) \quad (24)$$

The solution to the GBM process, is the most popular and wide used method to estimate stock price numerically. This is because under this process, unlike under an ABM a GBM induces non-zero strictly positive values. It is important to note this method is parametric and makes a Gaussian returns assumption, that is $\log(S_t/S_0) \sim N(\cdot)$, which can bias our estimates as heavy-tailed non-Gaussian returns are a stylised fact of financial markets.

5 Black and Scholes

This is a continuous-time, continuous-value model, that uses SDE's to define the expected discounted value of an option payoff.

5.1 Deriving the Black and Scholes PDE

We start with a riskless portfolio

$$\Pi_T = x f(S_t, t) + \Delta S_t \quad (25)$$

Where $\Pi_T(t = T) = \Pi_0 e^{rT}$ is the expected payoff of the option at time T that grows deterministically according to the risk-free rate r . We denote the portfolio quantity of the derivative (option) as $f(S_t, t)$ as x . Similarly, Δ denotes the quantity of the stock S_t .

We can take differentials to define the differential form of the risk-free portfolio; doing so we get

$$d\Pi_T = x df(S_t, t) + \Delta dS_t \quad (26)$$

Where dS_t follows an SDE identical to 18, solving this SDE, and enforcing the risk-free constraint will help us arrive at the Black-Scholes PDE.

$$\frac{d\Pi_T}{dt} = r\Pi_0 e^{rt} \quad (27)$$

$$d\Pi_T = r\Pi_T dt \quad (28)$$

so we now have

$$r\Pi_T dt = x df(S_t, t) + \Delta dS_t \quad (29)$$

The first term is a differential of the risk-free asset, $f(S_t, t)$, which is given by Ito's lemma. Letting $f = f(S_t, t)$

$$\begin{aligned} r\Pi_T dt &= x \left(\left(f_t + \mu S_t f_{S_t} + \frac{1}{2}(\sigma S_t)^2 f_{S_t S_t} \right) dt + \sigma S_t f_{S_t} dB_t \right) + \Delta(\mu S_t dt + \sigma S_t dB_t) \\ &= x \left(f_t + \mu S_t f_{S_t} + \frac{1}{2}(\sigma S_t)^2 f_{S_t S_t} + \frac{1}{x} \Delta \mu S_t \right) dt + (\Delta \sigma S_t + x \sigma S_t f_{S_t}) dB_t \end{aligned} \quad (30)$$

For the portfolio to be risk-free, the diffusion term, $dB(t)$, must be zero.

$$\begin{aligned}
\Delta\sigma S_t + x\sigma S_t f_{S_t} &= 0 \\
\Delta &= -x f_{S_t} \\
&= -x \frac{\partial f(S_t, t)}{\partial S_t}
\end{aligned} \tag{31}$$

Equation 31 tells us we must invest a weighted quantity of the risk-free asset in order to cancel out the portfolio expected volatility. The weighted value, is the quantity of shares invested in the derivative and the weight is the sensitivity of the derivative with respect to a change in the underlying i.e. the immediate value the derivative changes by for a given change in the asset price. Letting $x = 1$ (one share of the asset) and substituting in the “portfolio delta” into 30

$$\begin{aligned}
r\Pi_T &= f_t + \mu S_t f_{S_t} + \frac{1}{2}(\sigma S_t)^2 f_{S_t S_t} - \mu S_t f_{S_t} \\
0 &= f_t + \frac{1}{2}\sigma^2 S_t^2 f_{S_t S_t} - r\Pi_T
\end{aligned}$$

Substituting in the definition of the portfolio payoff using equations 25 and 31, $\Pi_T = f(S_t, t) - S_t f_s$.

$$\frac{1}{2}(\sigma^2 S_t^2) \frac{\partial^2 f(S_t, t)}{\partial S_t^2} + \frac{\partial f(S_t, t)}{\partial t} + r \left(S_t \frac{\partial f(S_t, t)}{\partial S_t} - f(S_t, t) \right) = 0 \tag{32}$$

Equation 32 is the famous Black and Scholes partial differential equation (PDE).

5.2 Solving the Black and Scholes PDE

5.2.1 Change of Variables

First we apply a change of variables; we let $X_t = \log(S_t)$. We can then redefine our value function, given $S_t = e^{X_t}$ we let $f(S_t, t) = f(e^{X_t}, t) = u(X_t, t)$. The reasoning for applying this change of variables is so we can transform the Black-Scholes PDE into a constant coefficient PDE, which is easier to solve for analytically.

$$\frac{\partial x}{\partial s} = \frac{1}{s} \frac{\partial s}{\partial s} = \frac{1}{s} \tag{33}$$

Where $s = S_t$ and $x = X_t$. To apply the change of variables to our Black-Scholes PDE, 32, we must solve for the first and second order partials of the derivative, which we call the value function, with respect to the underlying, S_t .

Solving for $u_s = f_s$,

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial s} = \frac{1}{s} \frac{\partial u}{\partial x} \tag{34}$$

Solving for the second partial $u_{ss} = f_{ss}$, we get

$$\begin{aligned}
\frac{\partial^2 f}{\partial s^2} &= \frac{\partial}{\partial s} \left(\frac{1}{s} \frac{\partial u}{\partial x} \right) \\
&= \frac{\partial u}{\partial x} \frac{\partial}{\partial s} \left(\frac{1}{s} \right) + \frac{1}{s} \left(\frac{\partial u_x}{\partial x} \frac{\partial x}{\partial s} \right) \\
&= -\frac{1}{s^2} \frac{\partial u}{\partial x} + \frac{1}{s^2} \frac{\partial^2 u}{\partial x^2} \\
&= \frac{1}{s^2} (u_{xx} - u_x)
\end{aligned} \tag{35}$$

Given our change of variables is only for X_t , the value function derivative under the change of variables is just u_t . Substituting 34 and 35 into the black and scholes PDE, 32, we get

$$\frac{1}{2}(\sigma s)^2 \frac{1}{s^2}(u_{xx} - u_x) + u_t + r(s \frac{1}{s}u_x - u) = 0 \quad (36)$$

Grouping by order of partials, we get the Black and Scholes PDE under the change of variables, 37.

$$\frac{\sigma^2}{2}u_{xx} + \left(r - \frac{\sigma^2}{2}\right)u_x + u_t - ru = 0 \quad (37)$$

Notice how the problem now does not include S_t coefficients making the problem easier to solve.

5.2.2 The Heat Equation

We now apply a function transformation to 38 to get a PDE proportional to the classical heat equation, which we know how to solve. We can then simply take the inverse transform (as unlike a change of variables, this transformation is not equivalent) to arrive at the final solution to the Black-Scholes PDE.

We apply an exponential transformation, specifically

$$u(x, t) = e^{Ax+Bt}v(x, t) \quad (38)$$

Equation 38 introduces terms A and B that solve to cancel the v_x term. Solving for v_t

$$\begin{aligned} u_t &= \frac{\partial}{\partial t}(e^{Ax+Bt}v(x, t)) \\ &= Be^{Ax+Bt}v + e^{Ax+Bt}v_t \\ &= e^{Ax+Bt}(Bv + v_t) \end{aligned} \quad (39)$$

Solving for u_x

$$\begin{aligned} u_x &= \frac{\partial}{\partial x}e^{Ax+Bt}v(x, t) \\ &= Ae^{Ax+Bt}v + e^{Ax+Bt}v_x \\ &= e^{Ax+Bt}(Av + v_x) \end{aligned} \quad (40)$$

And finally, solving for the second derivative of the value function with respect to the underlying.

$$\begin{aligned} u_{xx} &= \frac{\partial}{\partial x}u_x \\ &= (A^2e^{Ax+Bt}v + Ae^{Ax+Bt}v_x) + (Ae^{Ax+Bt}v_x + e^{Ax+Bt}v_{xx}) \\ &= e^{Ax+Bt}(A^2v + 2Av_x + v_{xx}) \end{aligned} \quad (41)$$

Substituting 39, 40 and 41 and into 38 we get

$$\begin{aligned} \frac{\sigma^2}{2} e^{Ax+Bt} (A^2 v + 2A v_x + v_{xx}) + \left(r - \frac{\sigma^2}{2}\right) e^{Ax+Bt} (A v + v_x) + e^{Ax+Bt} (B v + v_t) - r e^{Ax+Bt} v &= 0 \\ e^{Ax+Bt} (v_t + \frac{\sigma^2}{2} v_{xx}) + e^{Ax+Bt} ((B + \frac{\sigma^2}{2} A^2 + (r - \frac{\sigma^2}{2}) A - r) v + (\sigma^2 A - \frac{\sigma^2}{2} + r) v_x) &= 0 \end{aligned} \quad (42)$$

Solving for A

$$\begin{aligned} \frac{\sigma^2}{2} A + r - \frac{\sigma^2}{2} &= 0 \\ A &= \frac{1}{2} - \frac{r}{\sigma^2} \end{aligned} \quad (43)$$

Substituting A into the v_x coefficient, [42](#), drops it out the equation (it equals zero). The v coefficient remains. We can now solve for the last unknown B

$$B = -\frac{\sigma^2}{2} \left(\frac{1}{2} - \frac{r}{\sigma^2}\right)^2 - \left(r - \frac{\sigma^2}{2}\right) \left(\frac{1}{2} - \frac{r}{\sigma^2}\right) + r \quad (44)$$

Let $\frac{\sigma^2}{2} A^2 + (r - \frac{\sigma^2}{2}) A - r = c$ then $B = -c$ and therefore $c + (-c) = 0$ so both coefficients evaluate to zero and we are left with.

$$e^{Ax+Bt} \left(v_t + \frac{\sigma^2}{2} v_{xx} \right) = 0 \quad (45)$$

Focusing on the inner term, this is identical to the heat equation. We can now focus on solving this PDE, which will give us all the required information to solve the original Black-Scholes PDE for some value function v .

$$v_t = -\frac{\sigma^2}{2} v_{xx} \quad (46)$$

5.2.3 Time Reversal

Equation [46](#) is the backward heat equation that describes the evolution of a process from maturity to today; we need a problem that describes evolution from today into the future. Specifically, we let $\tau = T - t$. τ is referred to as the time until maturity of the derivative (option) contract; $t : T \rightarrow 0$ and $\tau : 0 \rightarrow T$ so τ is forward in time and t is backward in calendar time.

$$\begin{aligned} v_t(x, \tau(t)) &= \frac{\partial v}{\partial \tau} \frac{\partial \tau}{\partial t} \\ &= -v_\tau \end{aligned} \quad (47)$$

Making the substitution, the classical forward heat equation that describes the evolution of some multivariate function from now into the future.

$$v_\tau = \frac{\sigma^2}{2} v_{xx} \quad (48)$$

Importantly as a direct consequence of the $v_t = v_\tau$, the B term is now multiplied by negative

$$B = \frac{\sigma^2}{2}A^2 + \left(r - \frac{\sigma^2}{2}\right)A - r$$

Solving 48, applying the inverse transformation, and the option maturity boundary condition will give us the derivative (option) payoff t days from the present.

5.2.4 Fourier Series and Fourier Transformation

Equation 48 still remains a PDE, although now much easier to solve than in 32. By applying a Fourier transform to each partial term we can reduce the PDE to an ODE, then solve the ODE in frequency space. This transformation is invertible, so we can then apply an inverse Fourier transform to get the original solution in feature space.

A Fourier transform of a function $f(x)$ is

$$\begin{aligned}\mathcal{F}(f(x)) &= \int_{-\infty}^{\infty} f(x)e^{i\omega x}dx \\ &= \hat{f}(\omega)\end{aligned}\tag{49}$$

Where $\omega = \frac{k\pi}{L}$, The inverse Fourier transform is

$$\begin{aligned}\mathcal{F}^{-1}(\hat{f}(\omega)) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega)e^{i\omega x}d\omega \\ &= f(x)\end{aligned}\tag{50}$$

We will also note some useful properties of the Fourier transform such as the invariance of the derivative of the Fourier transform with respect to x .

$$\frac{d^n}{dx^n}\mathcal{F}(f(x)) = (i\omega)^n \hat{f}(x)\tag{51}$$

To transform 48 into the frequency domain we apply the Fourier transform, 49, to the portfolio value function $v(x, \tau)$ which we denote $\hat{v}(\omega, \tau)$. Using equation 51

$$\hat{v}_\omega = i\omega \hat{v}\tag{52}$$

$$\hat{v}_{\omega\omega} = -\omega^2 \hat{v}\tag{53}$$

And we denote $\hat{v} = \hat{v}(\omega, \tau)$

5.2.5 Frequency Domain ODE and Inverse Fourier Transform

Substituting the spectral derivatives of the portfolio value function, we have

$$\frac{d}{d\tau}\hat{v} + \frac{\sigma^2}{2}\omega^2\hat{v} = 0\tag{54}$$

Equation 54 is a ODE of form $\frac{d}{dx}(f(x)) + P(x)f(x) = Q(x)$, which can be solved by using the integrating factor where $P(\tau) = \frac{1}{2}\sigma^2\omega^2 \implies \tau = 1$.

$$\exp\left\{\int P(\tau)d\tau\right\}\left(\frac{d\hat{f}(\tau)}{d\tau} + P(\tau)\hat{v}\right) = 0\tag{55}$$

$$\implies \frac{d}{d\tau}\left(\exp\left\{\int P(\tau)d\tau\right\}\hat{v}\right) = 0d\tau\tag{56}$$

Solving equations 55 and 55 will yield the solution to $\hat{v}$. First, solving the integrating factor

$$\int P(\tau) d\tau = \frac{\sigma^2}{2} \omega^2 \int d\tau = \frac{\sigma^2}{2} \omega^2 \tau \quad (57)$$

Then substituting in for 55 we have

$$e^{\frac{\sigma^2}{2} \omega^2 \tau} \left(\hat{v}' + \frac{\sigma^2}{2} \omega^2 \hat{v} \right) = 0 \quad (58)$$

Substituting in for 56

$$\begin{aligned} \frac{d}{d\tau} \left(e^{\frac{\sigma^2}{2} \omega^2 \tau} \hat{v} \right) &= 0 \\ e^{\frac{\sigma^2}{2} \omega^2 \tau} \hat{v} &= \hat{v}(\omega, \tau = 0) \\ \hat{v} &= e^{-\frac{\sigma^2}{2} \omega^2 \tau} \hat{v}(\omega, \tau = 0) \end{aligned} \quad (59)$$

Equation 59 is the heat equation solution in frequency space; to get the solution in feature space, we must apply an inverse Fourier transform. We will also use a property of the Fourier transform to reformulate the calculation of a product of two transformed functions of in the same variable.

$$\mathcal{F}(f(x)g(x)) = \mathcal{F}(f(x)) * \mathcal{F}(g(x)) = \hat{f}(x)\hat{g}(x) \quad (60)$$

Equation 60 tells us the product of two Fourier-transformed functions is identical to the convolution of the Fourier transform of these functions, assuming identical variables. The convolution is a way to calculate the sum of two functions and is given as follows

$$(\mathcal{F}(f) * \mathcal{F}(g))(x) = \int \mathcal{F}(f(y)) \mathcal{F}(g(x - y)) dy \quad (61)$$

Using 60 to simplify 61

$$\begin{aligned} \mathcal{F}^{-1}(\hat{v}) &= \mathcal{F}^{-1}(e^{-\frac{\sigma^2}{2} \omega^2 \tau} \hat{v}(\omega, \tau = 0)) \\ v(x, \tau) &= \mathcal{F}^{-1}(e^{-\frac{\sigma^2}{2} \omega^2 \tau}) * \mathcal{F}^{-1}(\hat{v}(\omega, \tau = 0)) \\ &= \mathcal{F}^{-1}(e^{-\frac{\sigma^2}{2} \omega^2 \tau}) * v(x, \tau = 0) \end{aligned} \quad (62)$$

Focusing on solving the first term

$$\begin{aligned} \mathcal{F}^{-1}(e^{-\frac{\sigma^2}{2} \omega^2 \tau}) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(\sigma\omega)^2 \tau} e^{i\omega x} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{\sigma^2}{2} \tau \omega^2 + i x \omega} d\omega \end{aligned} \quad (63)$$

Equation ?? contains a quadratic exponent in ω , and we can solve the integral by first completing the square.

$$a = \frac{1}{2} \sigma^2 \tau \implies -a\omega^2 + ix\omega = 0 \quad (64)$$

Completing the square

$$-a \left(\left(\omega + \frac{ix}{2a} \right)^2 - \left(\frac{ix}{2(-a)} \right)^2 \right) = -a \left(\omega + \frac{ix}{2a} \right)^2 - \frac{x^2}{4a} \quad (65)$$

Substituting 65 in 63

$$\begin{aligned} \mathcal{F}^{-1}(e^{-\frac{\sigma^2}{2}\omega^2\tau}) &= \frac{1}{2\pi} e^{-\frac{x^2}{4a}} \int_{-\infty}^{\infty} e^{-a(\omega + \frac{ix}{2a})^2} d\omega; \text{ let } c = \frac{ix}{2a} \\ &= \frac{1}{2\pi} e^{-\frac{x^2}{4a}} \int_{-\infty}^{\infty} e^{-a(\omega+c)^2} d\omega; \text{ let } u^2 = a(\omega+c)^2 \iff u = \sqrt{a}(\omega+c), d\omega = \frac{1}{\sqrt{a}} du \\ &= \left(\frac{1}{2\pi} e^{-\frac{x^2}{4a}} \frac{1}{\sqrt{a}} \right) \int_{-\infty}^{\infty} e^{-u^2} du \\ &= \left(\frac{1}{2\pi} e^{-\frac{x^2}{4a}} \frac{1}{\sqrt{a}} \right) \sqrt{\pi} \\ &= \frac{1}{2\pi} \sqrt{\frac{\pi}{a}} e^{-\frac{x^2}{4a}} \\ &= \frac{1}{2\sqrt{\pi}} \sqrt{\frac{2}{\sigma^2\tau}} e^{-\frac{x^2}{4} \frac{2}{\sigma^2\tau}} \end{aligned} \quad (66)$$

5.2.6 Black and Scholes Option Price Solution

Substituting the solution of the inverse Fourier transform into 63, further simplifying and using our transformation identities, we arrive at the solution to the original Black and Scholes PDE defined in ??

$$f(S_t, t) = u(\log(S_t), \tau = T - t) = e^{A \log(S_t) + B(T-t)} v(x, \tau) \quad (67)$$

$$v(x, \tau) = \frac{1}{\sigma\sqrt{2\pi\tau}} e^{-\frac{x^2}{2\sigma^2\tau}} * v(x, \tau = 0) \quad (68)$$

$$v(x, \tau = 0) = e^{-Ax} u(x, 0) \quad (69)$$

The table below summarises the different components of the solution to the Black and Scholes PDE defined above.

Variables	Interpretation
$f(S_t, t)$	Black Scholes predicted option price
$u(x, \tau)$	Transformed price, $X_t = \log(S_t)$, and reverse time, $\tau = T - t$, solution to the Black and Scholes PDE
$v(x, \tau)$	Heat equation solution of transformed Black and Scholes problem
$v(x, \tau = 0)$	Transformed option payoff at maturity

5.3 Option Payoffs

To derive the Black and Scholes solution for simple combinations of long or short, call or put options, we can use the table below to determine the option payoff functions.

	call	put
long	$\max\{S_t - K, 0\}$	$\max\{K - S_t, 0\}$
short	$\min\{K - S_t, 0\}$	$\min\{S_t - K, 0\}$

Table 1: Option payoffs for long and short call and put options. Under Black and Scholes, we replace S_t with e^{X_t} .

We usually deal with long options as we are usually the buyer of the contract. For shorts, we are the seller of the contract.

5.4 Option Pricing under the Black and Scholes Model

Equation 67 is the general Black and Scholes option price solution for an arbitrary derivative (option), $v(x, \tau)$. In order to recover the option price under the Black and Scholes model, 67, we must define an option payoff at maturity and solve the option convolution, 62

5.4.1 European Call Option Solution

The payoff of a standard European call option –a call option that gives you the right but not the obligation to exercise the option at a price S_t at maturity, $\tau = 0$ – is given by equation 70

$$\Pi_c(S_T, K) = \max\{S_T - K, 0\} \quad (70)$$

The variable we convolve the two functions over is $x = X_t$ and we introduce a dummy variable y , which we integrate over. We will treat the option value at maturity as the f function in convolution, so $S_t = e^x = e^y$ under convolution and the exponential term will be the only function that is a function of x

$$v(x, \tau) = \frac{1}{\sigma\sqrt{2\pi\tau}} \int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2\tau}(x-y)^2} e^{-Ay} \max\{e^y - K, 0\} dy \quad (71)$$

To evaluate the integral we must define the maximum function piecewise and in y space (the log-space) as we are integrating over y .

$$\max\{e^y - K\} = \begin{cases} 0 & , \text{if } y - \log(K) \leq 0 \\ e^y - K & , \text{if } y - \log(K) > 0 \end{cases} \quad (72)$$

Now we can redefine our symmetric integral 71

$$\begin{aligned} v_C(x, \tau) &= \frac{1}{\sigma\sqrt{2\pi\tau}} \int_{\log(K)}^{\infty} e^{-\frac{1}{2\sigma^2\tau}(x-y)^2} e^{-Ay} (e^y - K) dy + \frac{1}{\sigma\sqrt{2\pi\tau}} \int_{-\infty}^{\log(K)} e^{-\frac{1}{2\sigma^2\tau}(x-y)^2} dy \\ &= \frac{1}{\sigma\sqrt{2\pi\tau}} \int_{\log(K)}^{\infty} e^{-\frac{1}{2\sigma^2\tau}(x-y)^2} e^{-Ay} (e^y - K) dy \\ &= \frac{1}{\sigma\sqrt{2\pi\tau}} \int_{\log(K)}^{\infty} e^{-\frac{1}{2\sigma^2\tau}(x-y)^2 + y - y(A-1)} dy - \frac{K}{\sigma\sqrt{2\pi\tau}} \int_{\log(K)}^{\infty} e^{-\frac{1}{2\sigma^2\tau}(x-y)^2 - Ay} dy \\ &= I_1 - I_2 \end{aligned} \quad (73)$$

Focusing on the first integral, I_1

$$\begin{aligned} a(x-y)^2 + y(A-1) &= -a \left(x^2 - 2xy + y^2 + y\frac{c}{a} \right), a = \frac{1}{2\sigma^2\tau}, c = A-1 \\ &= -a \left(y^2 - \frac{2xa-c}{a}y + x^2 \right), \text{let } \mu_1 = \frac{2xa-c}{2a} \\ &= -a((y-\mu_1)^2 - \mu_1^2 + x^2), \\ &= -a(y-\mu_1)^2 + a\mu_1^2 - ax^2 \end{aligned} \quad (74)$$

Substituting the completed square in the first

$$I_1 = \frac{1}{\sigma\sqrt{2\pi\tau}} \int_{\log(K)}^{\infty} e^{-a(y-\mu_1)^2 + a\mu_1^2 - ax^2} dy = \frac{1}{\sigma\sqrt{2\pi\tau}} e^{a(\mu_1^2 - x^2)} \int_{\log(K)}^{\infty} e^{-\frac{1}{2} \frac{1}{\sigma^2\tau} (y-\mu_1)^2} dy \quad (75)$$

Making a z substitution as follows

$$z = \frac{y - \mu_1}{\sigma\sqrt{\tau}}; dy = \sigma\sqrt{\tau}dz \quad (76)$$

And accounting for the limit of integration change under the change of variables

$$z(\log(K)) = \frac{\log(K) - \mu_1}{\sigma\sqrt{\tau}}; z(\infty) = \frac{\infty - \mu_1}{\sigma\sqrt{\tau}} = \infty \quad (77)$$

Substituting in 75

$$\begin{aligned} I_1 &= \frac{1}{\sigma\sqrt{2\pi\tau}} (\sigma\sqrt{\tau}) e^{-a(\mu_1^2 - x^2)} \int_{z(\log(K))}^{\infty} e^{-\frac{1}{2}z^2} dz = e^{a(\mu_1^2 - x^2)} \left(1 - \mathbb{P}\left(z < \frac{y - \mu_1}{\sigma\sqrt{\tau}}\right) \right) \\ &= e^{a(\mu_1^2 - x^2)} (1 - \Phi(z_1)); 1 - \Phi(a) = \Phi(-a) \\ &= e^{a(\mu_1^2 - x^2)} \Phi(-z_1) \end{aligned} \quad (78)$$

Starting with simplifying μ_1

$$\begin{aligned} \mu_1 &= x - \frac{1}{2a}c \\ &= x - \frac{1}{2a} \left(\frac{1}{2} - \frac{r}{\sigma^2} - 1 \right) \\ &= x - \frac{1}{2a} \left(\frac{\sigma^2 - 2r - 2\sigma^2}{2\sigma^2} \right) \\ &= x - \frac{1}{2} 2\sigma^2\tau \left(\frac{\sigma^2 - 2r - 2\sigma^2}{2\sigma^2} \right) \\ &= x - \sigma^2\tau \left(\frac{-\sigma^2 - 2r}{\sigma^2} \right) \\ &= x + \sigma^2\tau \left(\frac{\sigma^2 + 2r}{2\sigma^2} \right) \\ &= x + \tau \left(\frac{\sigma^2 + 2r}{2} \right) \\ &= x + \tau \left(r + \frac{1}{2}\sigma^2 \right) \end{aligned} \quad (79)$$

Substituting 79 in z and calling it z_1

$$\begin{aligned} z_1 &= \frac{\log(K) - \mu_1}{\sigma\sqrt{\tau}} \\ &= \frac{\log(K) - x - \tau \left(r + \frac{1}{2}\sigma^2 \right)}{\sigma\sqrt{\tau}} \\ \implies \Phi(-z_1) &= \Phi \left(\frac{\log\left(\frac{S_t}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)\tau}{\sigma\sqrt{\tau}} \right) = \Phi(d_1) \end{aligned} \quad (80)$$

Equation 80 is the famous first Gaussian integral Cumulative Distribution Function (CDF) value. To calculate the first integral exponential term we do

$$\begin{aligned}
e^{Ax+B\tau}e^{a(\mu_1^2-x^2)} &= e^{Ax+B\tau+a(\mu_1^2-x^2)} \\
&= \exp \left\{ \left(\frac{1}{2} - \frac{r}{\sigma^2} \right) x - \left(\frac{r + \frac{1}{2}\sigma^2}{2\sigma^2} \right) \tau + \frac{1}{2\sigma^2\tau} \left(\left(x + \left(r + \frac{\sigma^2}{2} \right) \tau \right)^2 - x^2 \right) \right\}; \alpha = r + \frac{\sigma^2}{2} \\
&= \exp \left\{ \left(\frac{1}{2} - \frac{r}{\sigma^2} \right) x - \left(\frac{r + \frac{1}{2}\sigma^2}{2\sigma^2} \right) \tau + \frac{1}{2\sigma^2\tau} (2x\alpha + \alpha^2) \right\} \\
&= \exp \left\{ \left(\frac{1}{2} - \frac{r}{\sigma^2} \right) x - \left(\frac{r + \frac{1}{2}\sigma^2}{2\sigma^2} \right) \tau + \frac{1}{\sigma^2\tau} \alpha + \frac{1}{2\sigma^2} \alpha^2 \right\} \\
&= \exp \left\{ \left(\frac{1}{2} - \frac{r}{\sigma^2} \right) x - \left(\frac{r + \frac{1}{2}\sigma^2}{2\sigma^2} \right) \tau + \left(\frac{r}{\sigma^2} + \frac{1}{2} \right) x + \left(\frac{r + \frac{1}{2}\sigma^2}{2\sigma^2} \right) \tau \right\} \\
&= \exp \left\{ \left(\frac{1}{2} - \frac{r}{\sigma^2} + \frac{1}{2} + \frac{r}{\sigma^2} \right) x \right\} \\
&= e^{\log(S_t)} \\
&= S_t
\end{aligned} \tag{81}$$

Combining 80 and 81 we can define the first of the two integrals to be

$$e^{Ax+B\tau}I_1 = S_t\Phi(d_1) \tag{82}$$

We use an identical method to solve for the second integral, complete the square, use a *z substitution*, simplify to obtain the Gaussian integral, and then transform back to feature space to obtain the solution under Black and Scholes. Our quadratic now has fewer terms, specifically

$$\begin{aligned}
-\frac{1}{2\sigma^2\tau}(x-y)^2 - Ay &= -a \left(x^2 - 2xy + y^2 + y\frac{A}{a} \right); \frac{1}{2\sigma^2\tau}, \frac{c}{a} = C \\
&= -a \left(y^2 - \frac{2xa - A}{a}y + x^2 \right), \text{ let } \mu_2 = \frac{2xa - c}{a} \\
&= -a((y - \mu_2)^2 - \mu_2^2 + x^2) \\
&= -a(y - \mu_2)^2 + a\mu_2^2 - ax^2
\end{aligned} \tag{83}$$

Making the substitution to get the integral in Gaussian form

$$z = \frac{y - \mu_2}{\sigma\sqrt{\tau}}; dy = \sigma\sqrt{\tau}dz \tag{84}$$

We must also account for the change in integration limits under the *z* substitution

$$z(\log(K)) = \frac{\log(K) - \mu_2}{\sigma\sqrt{\tau}}; z(\infty) = \frac{\infty - \mu_2}{\sigma\sqrt{\tau}} = \infty \tag{85}$$

Substituting transformations we can reformulate the second integral

$$\begin{aligned}
I_2 &= \frac{K}{\sigma\sqrt{2\pi\tau}}(-\sigma\sqrt{\tau}) \int_{z(\log(K))}^{\infty} e^{-\frac{1}{2}z^2 + a\mu_2^2 - ax^2} dz = -Ke^{a(\mu_2^2 - x^2)} \left(\frac{1}{\sqrt{2\pi}} \int_{z(\log(K))}^{\infty} e^{-\frac{1}{2}z^2} dz \right) \\
&= -Ke^{a(\mu_2^2 - x^2)} \left(1 - \mathbb{P} \left(z \leq \frac{y - \mu_2}{\sigma\sqrt{\tau}} \right) \right) \\
&= -Ke^{a(\mu_2^2 - x^2)} \Phi(-z_2)
\end{aligned} \tag{86}$$

Following an identical procedure to arrive at the solution we start by simplifying μ_2

$$\begin{aligned}
\mu_2 &= 2x - \frac{A}{a} \\
&= 2x - 2\sigma^2\tau \left(\frac{1}{2} - \frac{r}{\sigma^2} \right) \\
&= 2x - (\sigma^2\tau - 2\tau r) \\
&= x - \frac{1}{2}\sigma^2\tau + \tau r \\
&= x + \left(r - \frac{1}{2}\sigma^2 \right) \tau
\end{aligned} \tag{87}$$

Substituting in z_2 to uncover the CDF term

$$\begin{aligned}
z_2 &= \frac{\log(K) - \mu_2}{\sigma\sqrt{\tau}} \\
&= \frac{\log(K) - x - \left(r - \frac{1}{2}\sigma^2 \right) \tau}{\sigma\sqrt{\tau}} \\
\Rightarrow \Phi(-z_2) &= \Phi \left(\frac{\log \left(\frac{S_t}{K} \right) + \left(r - \frac{1}{2}\sigma^2 \right) \tau}{\sigma\sqrt{\tau}} \right) = \Phi(d_2)
\end{aligned} \tag{88}$$

Finally, computing the exponential term

$$\begin{aligned}
e^{Ax+B\tau} e^{a(\mu_1^2 - x^2)} &= e^{Ax+B\tau+a(\mu_1^2 - x^2)} \\
&= \exp \left\{ \left(\frac{1}{2} - \frac{r}{\sigma^2} \right) x - \left(\frac{r + \frac{1}{2}\sigma^2}{2\sigma^2} \right) \tau + \frac{1}{2\sigma^2\tau} \left(\left(x + \left(r - \frac{\sigma^2}{2} \right) \tau \right)^2 - x^2 \right) \right\}; \alpha = r - \frac{\sigma^2}{2} \\
&= \exp \left\{ \left(\frac{1}{2} - \frac{r}{\sigma^2} \right) x - \left(\frac{r + \frac{1}{2}\sigma^2}{2\sigma^2} \right) \tau + \frac{1}{2\sigma^2\tau} (2x\alpha + \alpha^2) \right\} \\
&= \exp \left\{ \left(\frac{1}{2} - \frac{r}{\sigma^2} \right) x - \left(\frac{r + \frac{1}{2}\sigma^2}{2\sigma^2} \right) \tau + \frac{1}{\sigma^2\tau} \alpha + \frac{1}{2\sigma^2} \alpha^2 \right\} \\
&= \exp \left\{ \left(\frac{1}{2} - \frac{r}{\sigma^2} \right) x - \left(\frac{r + \frac{1}{2}\sigma^2}{2\sigma^2} \right) \tau + \left(\frac{r}{\sigma^2} - \frac{1}{2} \right) x + \left(\frac{r - \frac{1}{2}\sigma^2}{2\sigma^2} \right) \tau \right\} \\
&= \exp \left\{ \left(\frac{1}{2} - \frac{r}{\sigma^2} - \frac{1}{2} + \frac{r}{\sigma^2} \right) x + \left(\frac{-r^2 - r\sigma^2 - \frac{1}{4}\sigma^4 + r^2 - r\sigma^2 + \frac{1}{4}\sigma^4}{2\sigma^2} \right) \tau \right\} \\
&= \exp \left\{ -\frac{2\sigma^2 r}{2\sigma^2} \tau \right\} \\
&= e^{-r\tau}
\end{aligned} \tag{89}$$

Combining solutions 88 and 89 we get

$$e^{Ax+B\tau} I_2 = e^{-r\tau} \Phi(d_2) \tag{90}$$

Using 82 and 90 we can finally obtain the call option price for a non-dividend-paying stock under the Black Scholes model

$$\begin{aligned}
C(S_t, t) &= u(x = \log(S_t), \tau = T - t) = e^{Ax+B\tau} v(x, \tau = 0) \\
&= S_t \Phi(d_1) - K e^{-r\tau} \Phi(d_2)
\end{aligned} \tag{91}$$

5.4.2 European Put Option Solution

Now that we have solved for a call option, we can use its solutions and quickly find the solution for a put option. A put option is derivative with payoff at maturity of

$$\Pi_P(S_T, K) = \max\{K - S_T, 0\} \quad (92)$$

Using identical reasoning, substituting in equation 68 we get

$$\frac{1}{\sigma\sqrt{2\pi\tau}} \int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2\tau}x^2} e^{-Ay} \max\{K - e^y, 0\} dy \quad (93)$$

Decomposing the maximum expression and re-defining the integration bounds we have

$$\begin{aligned} v_P(x, \tau) &= \frac{1}{\sigma\sqrt{2\pi\tau}} \int_{-\infty}^{\log(K)} e^{-\frac{1}{2\sigma^2\tau}(x-y)^2 - Ay} (K - e^y) dy \\ &= \frac{K}{\sigma\sqrt{2\pi\tau}} \int_{-\infty}^{\log(K)} e^{-\frac{1}{2\sigma^2\tau}(x-y)^2 - Ay} dy - \frac{1}{\sigma\sqrt{2\pi\tau}} \int_{-\infty}^{\log(K)} e^{-\frac{1}{2\sigma^2\tau}(x-y)^2 - (A-1)y} dy \end{aligned} \quad (94)$$

The only difference between the put option integral expression and the call integral expression from the heat equation is the integral sign and integral bounds. The first of the two integrals with reversed bounds is shown to $Ke^{-r\tau}\Phi(d_1)$, reversing the bounds means we are calculating the CDF for the left tail rather than the right, therefore the inverse CDF value will be the negative of the call, giving $Ke^{-r\tau}\Phi(-d_2)$. Following identical reasoning for the second integral we have $S_t\Phi(-d_1)$.

Therefore the Put option solution under Black and Scholes is given by equation 95

$$\begin{aligned} P(S_t, t) &= u(\log(S_t), \tau = T - t) = e^{Ax+B\tau} v(x, \tau = 0) \\ &= Ke^{-r\tau}\Phi(-d_2) - S_t\Phi(-d_1) \end{aligned} \quad (95)$$

5.5 Pricing Options in Practice under Black and Scholes

Example 1: Pricing a long call option under Black and Scholes

What is the option price under Black and Scholes at $t = 0$, given the option strike price $K = \$15$, the current asset price is $S_0 = 7$, a standard deviation of \$0.2 and a risk-free rate $r = 3\%$.

$$\begin{aligned} d_1 &= -3.56 \implies \Phi(-3.56) = 0.00019 \\ d_2 &= -3.76 \implies \Phi(-3.76) = 0.00008 \end{aligned}$$

So the discounted expected long call price of the European option at time $t = 0$ is

$$\delta = C(S_0, t = 0) = 30 \cdot 0.00019 - 15 \cdot e^{0.03} \cdot 0.00008 = \$0.0001645$$

5.6 Put-Call Parity

Put-call parity is an equivalence identity that tells us that an option contract's payoff can be replicated using a combination of holding a call or put contract, holding shares in the underlying and lending (borrowing) money to (from) the bank at the risk-free rate. Formally

$$P_t - C_t = Ke^{-r\tau} - S_t; \tau = T - t \quad (96)$$

Where P_t is the put option price at time t , C_t is the call option price at time t , K is the option strike price, $Ke^{-r\tau}$ is the amount borrowed from the bank at the risk-free rate r time from option expiry, and S_t is the underlying asset price.

The equality of these cash flows is independent of future behaviour of the stock and is model-independent. Importantly, put-call parity does not apply for American options (We are dealing exclusively with European options, so this is fine). If put-call parity does not hold an arbitrage opportunity exists, and we can make a risk-free profit by buying the underpriced portfolio and selling the overpriced portfolio such that our net position at $t = 0$ is zero. The value of our net portfolio will always be equal to the strike price K . At maturity time, T , we wind down the position by repaying / (withdrawing) our loan (money in our savings account), making an arbitrage (riskless) profit equal to $|K - B_0e^{rT}| > 0$.

5.6.1 Put Call Parity Under Black and Scholes

Using call and put option solutions 91 and 95 respectively we have

$$\begin{aligned} P(S_t, t) - C(S_t, t) &= Ke^{-r\tau}[\Phi(-d_2) + \Phi(d_2)] - S_t[\Phi(-d_1) + \Phi(d_1)] \\ &= Ke^{-r\tau} - S_t \end{aligned} \quad (97)$$

Equation ?? proves put call parity holds under Black-Scholes for plain vanilla European options.

5.7 Black and Scholes under Risk Neutral Pricing

Under risk-neutral pricing, the asset price grows according to the risk-free rate and the asset price today is the discounted expected payoff of the derivative (option) payoff at maturity. We use $\mathbb{Q}$ to denote the risk-neutral measure (An SDE with drift equal to the risk-free interest rate). The risk-neutral measure means our option price (an Ito-process) evolves according to the SDE $dS_t = rS_t dt + \sigma S_t dB(t)$.

5.7.1 Feynman-Kac Formulation

Formalising the statement above we have an equivalent formulation for solving for the option price under Black and Scholes (one may argue this solution is more elegant). This is called the Feynman-Kac solution to the Black and Scholes PDE. Feynman-Kac is an alternative method to solving for the option price $f(S_t, t)$ that satisfies the Black-Scholes PDE 32.

$$e^{-rT} \mathbb{E}^{\mathbb{Q}}[g(X_T) | S_t = S] = e^{-rT} G(K) \quad (98)$$

Where $X_t = e^N$ is the price of the underlying (e.g. stock) and $N \sim \mathcal{N}(\mu, T\sigma^2)$, $g(X_T)$ is the option payoff at maturity. $G(K)$ is the result for discounting the terminal option payoff under the risk-neutral measure. For plain vanilla long calls and puts we have

$$C(K) = \int_K^\infty (1 - F(x)) dx \quad (99)$$

$$P(K) = \int_0^K F(x) dx \quad (100)$$

$C(K)$ and $P(K)$ denote the results for the expectation of the terminal option payoff under $\mathbb{Q}$ for calls and puts, [2]. $F(x)$ is the CDF function of X , which, as we known, is log-normally distributed (The exponential of a Gaussian is log-normal).

$$F(x) = \frac{1}{x\sqrt{2\pi\sigma^2T}} \exp \left\{ \frac{1}{2\sigma^2T} \left((\log(x) - \log(S_0) - (r - \frac{1}{2}\sigma^2)T) \right)^2 \right\} \quad (101)$$

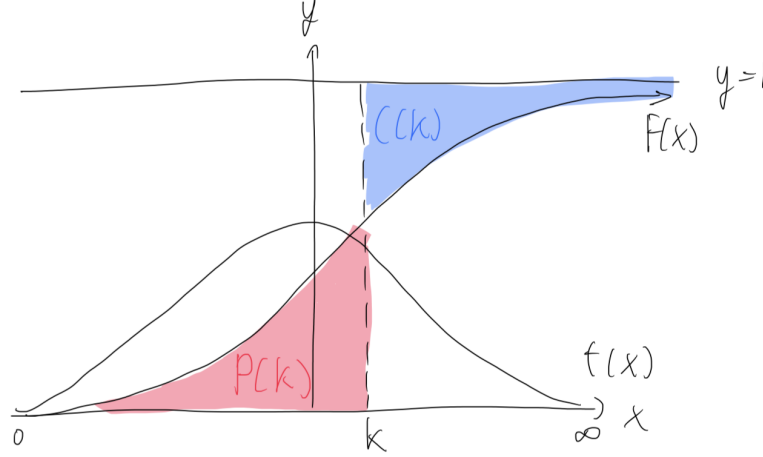


Figure 1: $f(x)$ and $F(x)$ are the probability density and cumulative probability density function of the underlying asset X which follow a log-normal density. The Blue shaded area is the call function and the red shaded area is the put function under the log-normal asset distribution (image by author).

Figure 1 helps represent the solution to the Black and Scholes PDE when using the risk-neutral measure solution. When using a risk-neutral measure, the Black and Scholes option price is exactly the discounted expected option payoff at maturity, which is effectively a scaled component of the bounded area above or below the CDF of the parametrised log-normal distribution. In the case of a call option, this would be the area $C(K)$ scaled by the discounting factor e^{-rT} .

We can evaluate the expectation in 98 for a call option to arrive at the Black and Scholes option pricing solution for a call option under risk neutral pricing.

$$\begin{aligned} e^{-rT} \mathbb{E}^{\mathbb{Q}}[g(X)] &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[\max\{e^N - K, 0\}] \\ &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[(e^N - K) \mathbb{1}\{N \geq \log(K)\}] \end{aligned} \quad (102)$$

To decompose this expectation, we used the following property of expectations

$$\mathbb{E}[g(X) \mathbb{1}\{x \geq a\}] = \int_a^\infty g(x) f_X(x) dx \quad (103)$$

Solving ?? will yield the discounted expected call option price under the Black and Scholes/the risk neutral measure $\mathbb{Q}$, identical to that shown in 91.

$$e^{-rT} \mathbb{E}^{\mathbb{Q}}[g(X)] = \frac{e^{-rT}}{\sigma\sqrt{2\pi T}} \int_{\log(K)}^\infty \exp \left\{ -\frac{1}{2\sigma^2T} \left((\log(\frac{x}{S_0}) - (r - \frac{1}{2}\sigma^2)T) \right)^2 \right\} (e^x - K) dx \quad (104)$$

Equation 104 is identical to solving the discounted integral of the underlying log-normal CDF function, $G(K) = C(K)$.

5.8 Option Premium and Option Payoff at Maturity

The premium, δ , refers to the discounted expected option payoff at maturity of the option under Black and Scholes, so basically the Black and Scholes prediction of option price at $t = 0$.

$$\delta = f(S_t, t = 0) \quad (105)$$

If we are long an option contract, we are paying the option premium, 105, with the hope of making at least the option premium back to realise a profit at $t = T$ (for European options). If we are shorting an option contract, we are selling/issuing the contract for the premium with the hope of retaining at most the premium to realise a profit. Formalising these statements, for a call we have

$$\Pi(S_T) = \begin{cases} \max\{S_T - K, 0\} - \delta \geq -\delta & , \text{if long call} \\ \min\{K - S_T, 0\} + \delta \leq \delta & , \text{if short call} \end{cases} \quad (106)$$

Similarly, for put option contracts, the respective payoffs at maturity are

$$\Pi(S_T) = \begin{cases} \max\{K - S_T, 0\} - \delta \geq -\delta & , \text{if long put} \\ \min\{S_T - K, 0\} + \delta \leq \delta & , \text{if short put} \end{cases} \quad (107)$$

The option payoffs at maturity for long and short, call and put options under Black and Scholes can be visualised graphically as follows.

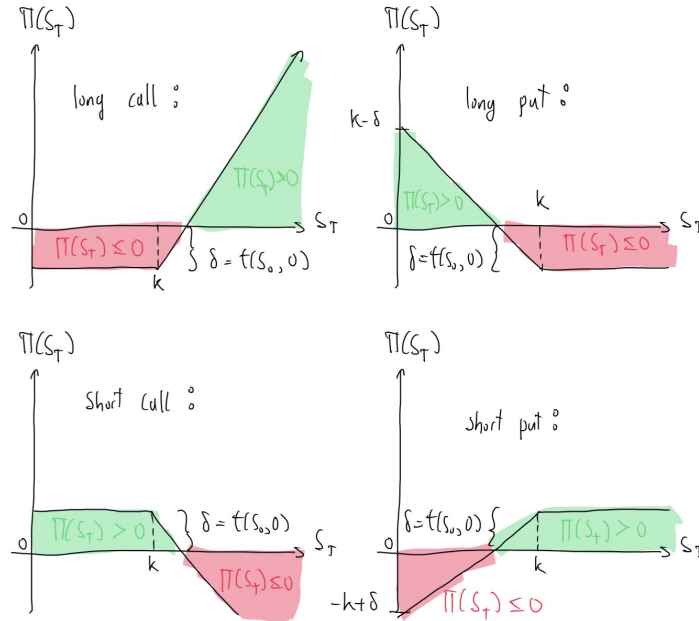


Figure 2: long call and long put, 106, (top row) and short call and short put 107, (bottom row) option payoffs at maturity with Black and Scholes option premiums (image by author)

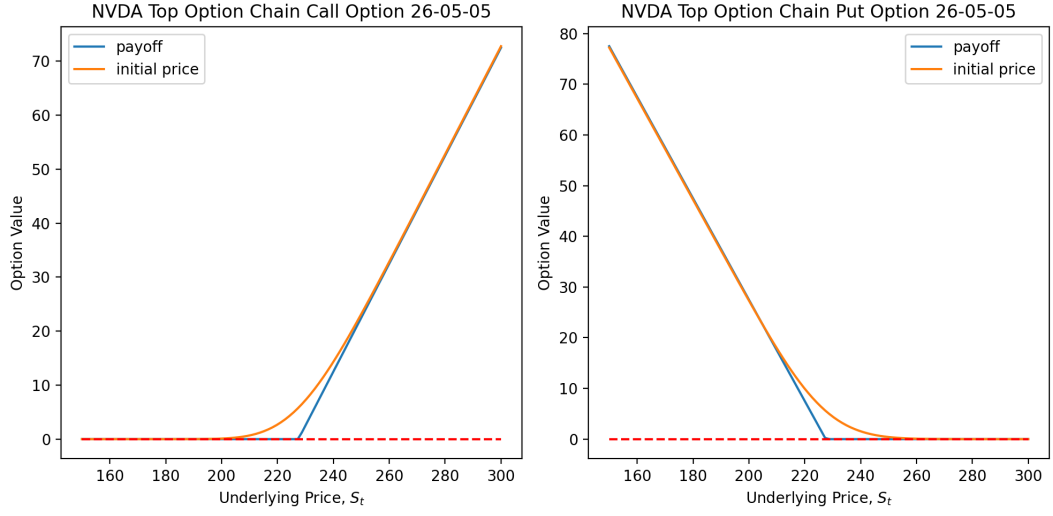


Figure 3: Top of Option Chain NVDA long Call (left) and long Put (right) Black and Scholes option predicted present value option price, $t = 0$, and option payoff at maturity. Both Call and Put contracts have an identical strike price, $K = \$227.5$. Both options were issued on 2026-05-05, and expire on 2026-05-13.

5.8.1 Option Premium Surface

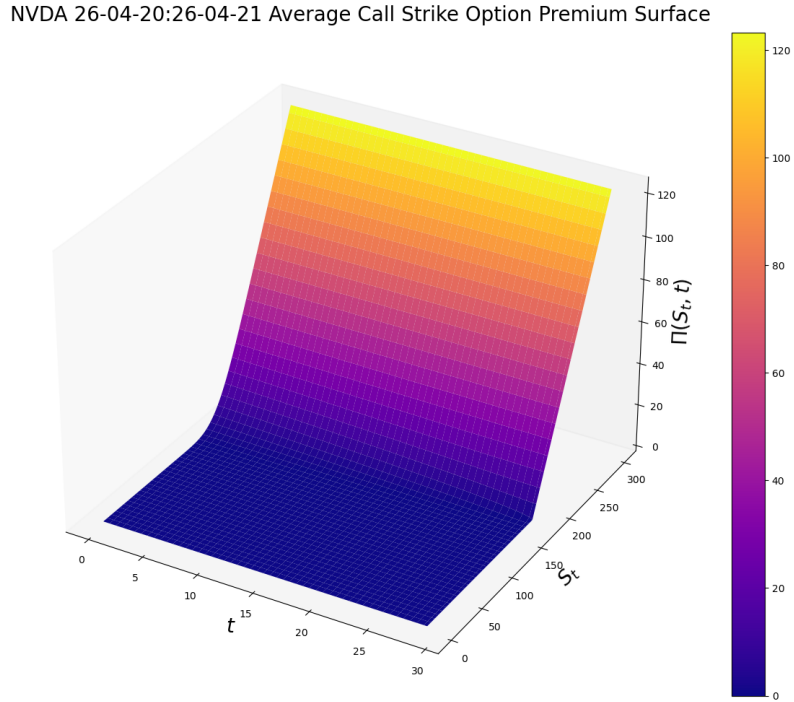


Figure 4: NVDA top-of-chain call option premium payoff surface through time. $K = \$227.5$, $t_0 = 2026-05-05$, $T = 2026-05-13$.

NVDA 26-04-20:26-04-21 Average Put Strike Option Premium Surface

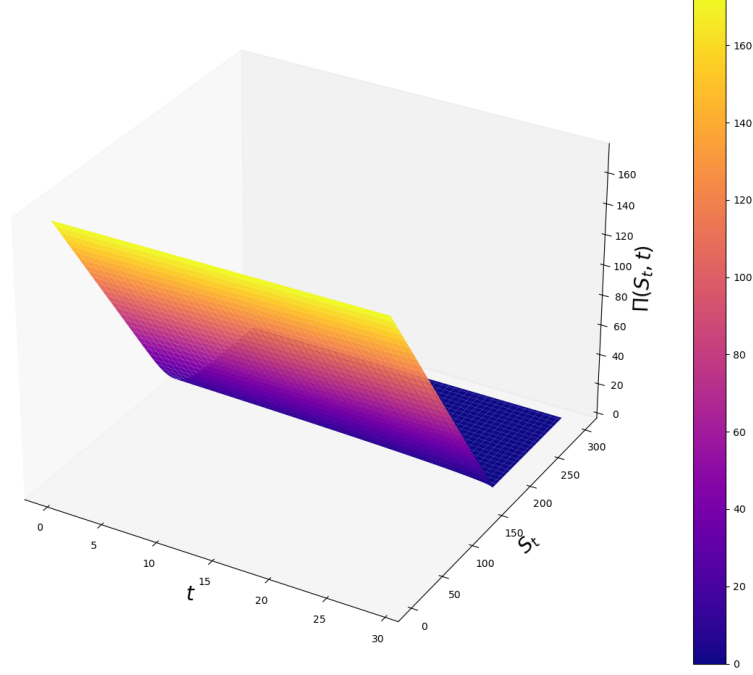


Figure 5: NVDA top-of-chain put option premium payoff surface through time. $K = \$227.5$, $t_0 = 2026-05-05$, $T = 2026-05-13$.

6 Volatility

Volatility is a measure of asset dispersion and play a crucial part in option pricing using Black and Scholes.

6.1 Realised Volatility (RV)

This is historical volatility, σ^2 , that tells us the degree to which the underlying asset's price moved in the past. We can easily estimate realised volatility using the unbiased variance.

$$\sigma^2 = \frac{1}{n-1} \sum_{i=1}^n (r_i - \mu)^2; r_i = \log \left(\frac{S_i}{S_{i-1}} \right) \quad (108)$$

6.2 Implied Volatility (IV)

Implied volatility (IV), $\tilde{\sigma}^2$, is a forward-looking estimate of volatility which, when used as a parameter in the Black-scholes model, should accurately approximate the market option price. IV tells what the market's future expectation of the option price is, if the market price is different from the option-predicted model price using realised volatility, a risky profit opportunity exists (as we will discuss shortly). We can easily solve for implied volatility using a grid search and stopping rule

$$\tilde{\sigma}^2 = \arg \min_{\sigma^2} \{ |f(S_t, t; r, \sigma^2, K) - f_m| \} \leq \varepsilon \quad (109)$$

Where $f(S_t, t; r, \sigma^2, K)$ is the option price under Black and Scholes and f_m is the observed option price. Equation ?? defines the loss function we minimise subject to a minimum

precision stopping rule. There are other ways to solve for implied volatility, although as with any brute-force algorithm, it is intuitive and has high correctness.

7 The Greeks

The Greeks are the first-order, second-order and higher order partial derivatives of the Black and Scholes option price for an option with respect to different underlying variables, commonly S_t, σ and t .

7.1 Delta

This is the partial derivative of the Black and Scholes option price with respect to the price of the underlying.

$$\Delta_t = \frac{\partial f(S_t, t)}{\partial S_t} \quad (110)$$

We encountered this exact quantity when deriving the Black–Scholes PDE, specifically when we were required to eliminate the diffusion term dB_t in our replicating portfolio under the risk-free assumption (see equation 31). We do this by holding a negative weighted quantity in the underlying asset; this quantity is the Δ (holding a negative quantity in the underlying is done by shorting the underlying). We interpret Δ as the instantaneous rate of change of the option price given a small change in the underlying, so it answers the question: if the underlying changes by some amount, how much does the option value change by?

The delta for a long call is

$$\begin{aligned} \Delta_C &= \frac{\partial}{\partial S_t}(S_t \Phi(d_1) - Ke^{-r\tau} \Phi(d_2)) \\ &= \Phi(d_1) + S_t \frac{\partial}{\partial S_t} \Phi(d_1) - Ke^{-r\tau} \frac{\partial}{\partial S_t} \Phi(d_2) \\ &= \Phi(d_1) - \frac{1}{\sigma\sqrt{\tau}} \left(\Phi'(d_1) - \frac{Ke^{-r\tau}}{S} \Phi'(d_2) \right) \end{aligned} \quad (111)$$

Equation 111 can be simplified by noting the identity

$$S\Phi'(d_1) = Ke^{-r\tau} \Phi'(d_2) \quad (112)$$

Substituting and simplifying:

$$\Delta_C = \Phi(d_1) \quad (113)$$

For a long call (short put) the call delta, Δ_C , is bounded between zero and one, $\Delta_C \in [0, 1]$. The delta for a long put is

$$\begin{aligned} \Delta_P &= \frac{\partial}{\partial S_t}(Ke^{-r\tau} \Phi(-d_2) - S_t \Phi(-d_1)) \\ &= -\Phi(-d_1) \\ &= -(1 - \Phi(d_1)) \\ &= \Phi(d_1) - 1 \\ &= \Delta_C - 1 \end{aligned} \quad (114)$$

For a long put (short call), the put delta, Δ_P , is bounded between negative one and zero, $\Delta_P \in [-1, 0]$.

The delta is very useful as it can be used for portfolio risk management. For example, if we are long one call option in stock ABC that controls 100 shares, and calculate the long call delta to be 0.5 (the option changes in value by 0.5 for a \$1 change in stock price), if we hold negative 50 shares (short 50 shares) of the underlying, we can maintain the same portfolio value no matter what direction the underlying takes.

The delta for a call and put is discontinuous at expiry. For an in-the-money (ITM) call option, $S_t > K$, as $t \rightarrow T$ the denominator of $d_1 \rightarrow \infty$ as $\sigma\sqrt{\tau} \rightarrow 0$. If the option is out-the-money (OTM), $S_t \leq K$ and the option expires worthless, so $\Delta_C = 0$ and $\Delta_P = -1$.

7.1.1 Delta Hedging

Delta hedging is a trading strategy that is normally used by institutional traders to offset adverse price movements by constructing a hedged portfolio.

If actual volatility, σ^2 , is greater than implied volatility $\tilde{\sigma}^2$, then the market is overpricing future volatility, so the option is underpriced. Taking advantage of the mispricing, we can long the option. Furthermore, we can apply a delta hedge strategy to lock in risk-free profit at expiry. We essentially build a portfolio, long an option, and short the underlying. Tracking the portfolio's evolution, we can prove the risk-free profit.

Components	Value
Option	V^i
Delta Hedge	$-\Delta S$
Cash Account	$\Delta S - V^i$

Table 2: Portfolio components

Components	Value
Option	$V^i + dV^i$
Delta Hedge	$-\Delta S - \Delta dS$
Cash Account	$(\Delta S - V^i)(1 + rdt)$

Table 3: Portfolio components post-investment

At t_0 , we long the option, paying the premium, $-V^i$, we then offset first-order changes in the option value wrt movements in the underlying by building the delta hedge and shorting the underlying, receiving ΔS_t in our cash account. The portfolio value today is $\Delta S - V^i$, which we invest in a savings account with growth rate r . The portfolio mark-to-market profit, $d\Pi$, is an infinitesimal change in the portfolio, using the terms in Table 3

$$\begin{aligned} d\Pi^i &= V^i + dV^i - (\Delta S + \Delta dS) + (\Delta S - V^i)(1 + rdt) \\ &= dV^i - \Delta S - r(V^i - \Delta S)dt \end{aligned} \tag{115}$$

If the option were correctly priced, we would have

$$d\Pi^a = dV^a - \Delta S - r(V^a - \Delta S)dt \tag{116}$$

The difference in mark-to-market profit over one infinitesimal time step is

$$\begin{aligned}
d\Pi^i - d\Pi^a &= dV^i - dV^a - r(V^i - V^a)dt, \text{ let } X_t = V^i - V^a \\
&= dX_t - rX_tdt, \text{ simplify by integrating factor, } \mu(x) = -r \int_0^t dt = -rt \\
&= e^{-rt}(dX_t - rX_tdt) \\
&= d(e^{-rt}X_t) \\
e^{rt}(d\Pi^i - d\Pi^a) &= e^{rt}d(e^{-rt}X_t) \\
&= e^{rt}d(e^{-rt}(V^i - V^a))
\end{aligned} \tag{117}$$

Equation 117 is the difference in the infinitesimal change in portfolio value (between the implied and actual option portfolios) at time t . Alternatively, it is the profit of the arbitrage opportunity at time t . We can get the present value of the profit of the arbitrage opportunity, 118, by discounting the profit at time t to t_0 .

$$e^{-r(t-t_0)}e^{rt}(d\Pi^i - d\Pi^a) = e^{-rt_0}d(e^{-rt}(V^i - V^a)) \tag{118}$$

We can now easily get the payoff at expiry time, 119, by summing over all times; in the limit, we integrate over time

$$\begin{aligned}
e^{-rt_0} \int_{t_0}^T d(e^{-rt}(V^i - V^a)) &= e^{rt_0}[e^{-rt}X_t]_{t_0}^T \\
&= V_{t_0}^a - V_{t_0}^i > 0
\end{aligned} \tag{119}$$

So the arbitrage profit is locked in at t_0 . Equation 119 tells us that if we buy a mispriced option and delta hedge continuously, the continuously discounted profit is equal to the original mispricing. Note this is an idealised assumption and assumes that our delta hedge perfectly replicates the option payoff such that our replicating portfolio grows exactly at the risk-free rate. So to summarise Black-Scholes

The one time step mark-to-market profit of our portfolio, Π^i is

$$\begin{aligned}
d\Pi^i &= dV^i - \Delta dS - r(V^i - \Delta S)dt, \text{ Using Ito's lemma to evaluate } dV^i \\
&= \left(\Theta^i dt + \Delta^i dS + \frac{1}{2}\sigma^2 S^2 \Gamma^i dt \right) - \Delta dS - r(V^i - \Delta S)dt \\
&= \Theta^i dt - r(V^i - \Delta S)dt + (\Delta^i - \Delta)dS + \frac{1}{2}\sigma^2 S^2 \Gamma^i dt, \text{ use BS PDE} \\
&= (\Delta^i - \Delta)dS + \frac{1}{2}(\sigma^2 - \tilde{\sigma}^2)S^2 \Gamma^i dt
\end{aligned} \tag{120}$$

Where Θ^i is the rate of change of option value with respect to time and Γ^i is the second-order rate of change of option value with respect to the underlying. Both terms will be covered later in this chapter.

7.1.2 Delta Hedging with Realised Volatility (RV)

We have been using the superscript, i , to denote implied volatility; we will use the superscript a to denote actual volatility. If we delta hedge with actual volatility, then from equation 120, the mark-to-market (MTM) profit will be a function of a stochastic and deterministic term. The randomness comes from the Brownian motion term, dB_t , in dS .

7.1.3 Delta Hedging with Implied Volatility (IV)

The option should be priced using implied volatility; therefore if we offset changes in the option with changes in the underlying using implied volatility we can drop the dS term, leaving a deterministic mark-to-market profit

$$d\Pi^i = \frac{1}{2}(\sigma^2 - \tilde{\sigma}^2)S^2\Gamma^i dt \quad (121)$$

We can sum up the present value of the mark-to-market profit of the portfolio to get the total profit of the mark-to-market portfolio.

$$\Pi_T^i = \int_{t_0}^T d\Pi_{t_0}^i = \frac{1}{2}(\sigma^2 - \tilde{\sigma}^2) \int_{t_0}^T e^{-r(t-t_0)} S^2 \Gamma^i dt \quad (122)$$

Because the stock price follows a log-normal random walk, the S term is a function of time despite dropping the time subscript, and is equal to the solution to a GBM given by equation 22. This integral is never computed in practice and rather estimated, summing over discrete time steps from t_0 to T .

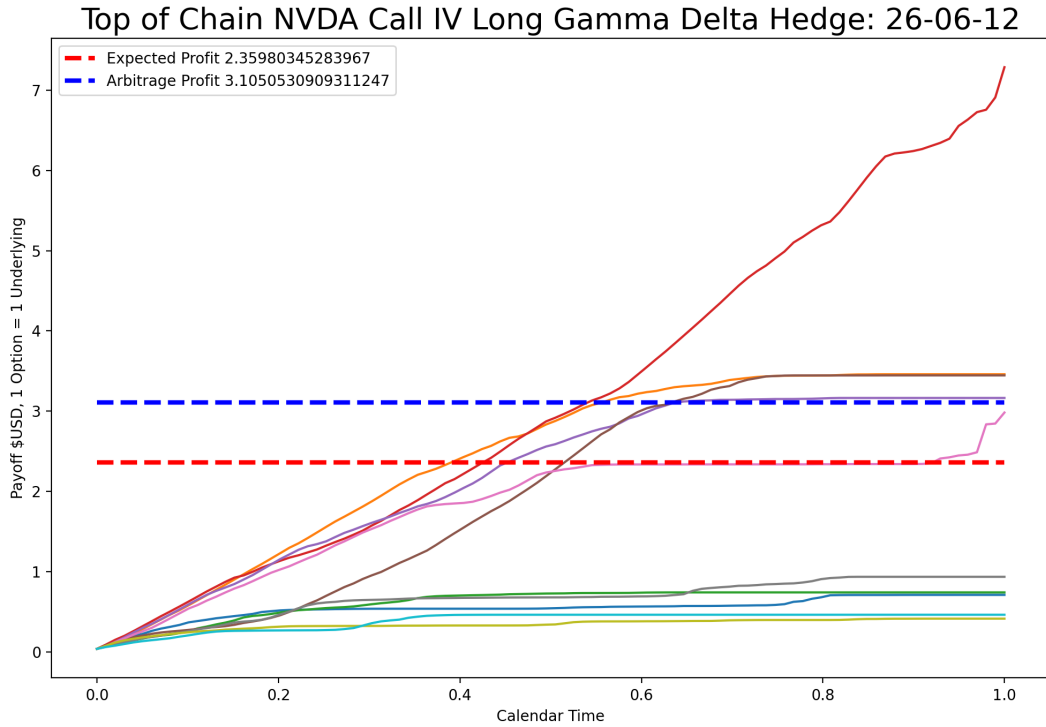


Figure 6: Total profit of delta hedged long gamma portfolio over 100 possible price paths, from option creation to expiry, assuming continuous delta hedge. Option type = C , $K = \$140.0$, $S_0 = \$130.0$, implied volatility = 0.19 option issued on 2026-05-05 and option expiry on 2026-07-31.

So to summarise, if an option is mispriced, $\tilde{\sigma}^2 \neq \sigma^2$, then an arbitrage opportunity exists. The black-scholes option payoff can be replicated by a perfect delta hedge, this means we delta hedge continuously (assuming no fees) so our position in the option replicates the option payoff. If we buy the mispriced option, delta hedging using implied volatility we have

essentially replicated a more expensive option more cheaply, our theoretical payoff will be equal to the initial option value mispricing, as noted by equation 117.

In practice we cannot delta hedge continuously, and our profit mark-to-market is a function of random components, S_t . In the limit of future paths the underlying can take to expiry, the expectation of our total profit should tend towards the initial mispricing [Figure 6].

7.1.4 Conditional Expected Profit of Long Gamma Delta IV Hedge

To get a better idea of what factors influence the profit of our implied delta hedge, we can evaluate the profit over a set of input parameters.

We will simulate the expected profit from a long-gamma (realised volatility greater than implied volatility) mispricing using an implied volatility delta hedge, given different underlying average returns and strike prices. We will proceed with deriving the continuous-time formulation of this conditional expected profit, which we will later approximate discretely.

$$\begin{aligned}\mathbb{E}[\Pi_T^i|\mu, K] &= \mathbb{E}\left[\int_{t_0}^T g(S_t, t; \mu) dt \middle| \mu, K\right], \text{using Fubini} \\ &= \int_{t_0}^T \mathbb{E}[g(S_t, t; \mu, K)] dt \\ &= \int_{t_0}^T e^{-r(t-t_0)} \mathbb{E}[S_t(t; \mu)^2 \Gamma(S_t; K, \tilde{\sigma}^2) | \mu, K] dt\end{aligned}\quad (123)$$

We proceed with simplifying the conditional expectation using LOTUS

$$\begin{aligned}\mathbb{E}[S^2 \Gamma | \mu, K] &= \int_0^\infty s^2 \Gamma^i p_{S_t|\mu, K}(s) ds \\ &= \frac{1}{\sigma \pi \sqrt{2t\tau}} \int_0^\infty \exp\left\{-\frac{1}{2t} \left(\frac{\log(\frac{S_t}{S_0}) - (\mu - \frac{1}{2}\sigma^2)t}{\sigma}\right)^2 - \frac{1}{2} \left(\frac{\log(\frac{S_t}{K}) + (\mu + \frac{1}{2}\tilde{\sigma}^2)\tau}{\tilde{\sigma}\sqrt{\tau}}\right)^2\right\} \\ &= \frac{1}{\sigma \pi \sqrt{2t\tau}} \int_0^\infty \exp\left\{\frac{1}{2t\sigma^2} \mu S_t - \frac{1}{2} d_1^2\right\}\end{aligned}\quad (124)$$

Substituting this expression into 123 we get the solution for continuous total profit of a delta hedge given some average constant growth factor, μ , and option strike price, K . Evaluating this expression over a set of strike prices and growth factors will trace out the expected profit surface, which is $\{\mathbb{E}[\Pi_T^i | \mu_i, K_j] : i = 1, \dots, m, ; j = 1, \dots, n\}$

NVDA Long-gamma IV Delta Hedge Profit Surface: 26-05-06

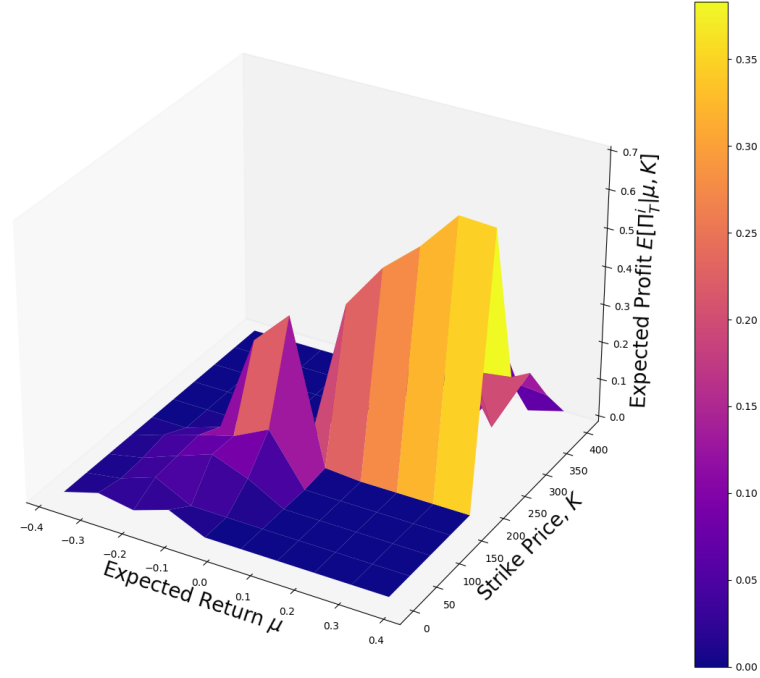


Figure 7: Empirical conditional expected profit surface of long-gamma implied volatility delta hedge. Profit conditioned on underlying return growth, μ , and option strike price, K . Portfolio is long-call and short underlying, $\tilde{\sigma} = 0.042$, $\sigma = 0.378$, historical returns 2025-05-07 to 2026-05-06.

NVDA Long-gamma IV Delta Hedge Profit Contour Map: 26-05-06

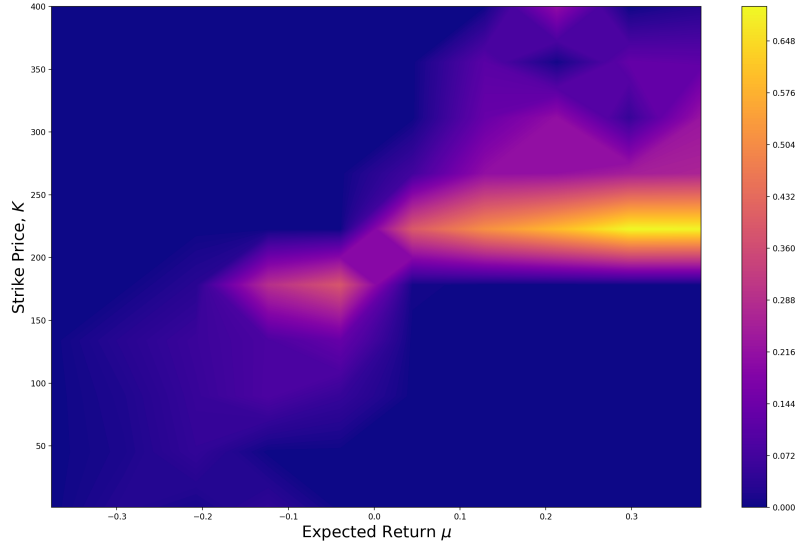


Figure 8: Empirical conditional expected profit contour map of long-gamma implied volatility delta hedge. Profit conditioned on underlying return growth, μ , and option strike price, K . Portfolio is long-call and short underlying, $\tilde{\sigma} = 0.042$, $\sigma = 0.378$, historical returns 2025-05-07 to 2026-05-06.

7.2 Gamma

Gamma measures the sensitivity of the change in option price with respect to changes in the underlying. It tells us how much the option delta changes. A high magnitude gamma means our option delta hedge is very responsive to changes in the underlying, requiring a frequent delta hedge rebalancing. The opposite is true for small-magnitude gamma. The gamma for a call and put are identical and strictly positive

$$\Gamma_C = \frac{\partial \Delta_C}{\partial S}, \Gamma_P = \frac{\partial \Delta_P}{\partial S} = \Gamma_C \quad (125)$$

Dropping the subscript and solving for the gamma of a call and put option we have

$$\begin{aligned} \Gamma &= \Phi'(d_1) \frac{\partial d_1}{\partial S} \\ &= \Phi'(d_1) \frac{1}{\sigma\sqrt{\tau}} \frac{\partial}{\partial S} (\log(S_t) - \log(K) + (\mu + \frac{1}{2}\sigma^2)\tau) \\ &= \frac{\Phi'(d_1)}{S_t\sigma\sqrt{\tau}} \end{aligned} \quad (126)$$

The second derivative measures the convexity of the option premium at any point in time. The gamma of a call and put option is greatest when the option is at-the-money (ATM), $S_t \approx K$. ATM $\log(S_t/K) \approx 0$ so $d_1 \approx 0$, given d_1 is standard normal and centered around zero, the PDF is maximised, giving the largest possible gamma. If however, the option is ITM or OTM as $t \rightarrow T$ the time to expiry tends to zero, $\sqrt{\tau} \rightarrow 0$, so $d_1 \rightarrow \pm\infty$ and $\Phi(d_1) \rightarrow 0$ so gamma is small.

7.2.1 Delta Gamma Hedging using Implied Volatility

We can improve our delta hedging strategy by hedging the second-order effects of the underlying in our portfolio. The gamma of the underlying is zero so we cannot add our delta hedge to eliminate the second-order effects of the stock price. Instead, we buy and sell another option, called the hedging option, in order to eliminate the volatility in our portfolio. We construct the following portfolio today

$$\Pi = V_1 + \lambda V_2 - \Delta S \quad (127)$$

Because the portfolio is delta and gamma neutral we have

$$\Pi_\Delta = 0 \implies \Delta = \Delta_1 + \lambda \Delta_2 \quad (128)$$

$$\Pi_\Gamma = 0 \implies \lambda = -\frac{\Gamma_1}{\Gamma_2} \quad (129)$$

Substituting the values of delta and gamma into the delta-gamma portfolio we have

$$\Pi = V_1 - \frac{\Gamma_1}{\Gamma_2} V_2 - (\Delta_1 + \lambda \Delta_2) S \quad (130)$$

To show the delta-gamma neutral strategy, we build a self-financing portfolio and track its evolution over time.

Components	Value
Option	$V_1^i + V_2^i$
Delta Hedge	$-\Delta S$
Cash Account	$\Delta S - V_1^i - V_2^i$

Table 4: Portfolio components

Components	Value
Option	$V_1^i + d_1 V^i + V_2^i + dV_2^i$
Delta Hedge	$-\Delta S - \Delta dS$
Cash Account	$(\Delta S - V_1^i - V_2^i)(1 + rdt)$

Table 5: Portfolio components post-investment

The mark-to-market profit of the delta-gamma neutral trading strategy is given by the sum of values in table 5

$$\begin{aligned}
d\Pi^i &= dV_1^i + dV_2^i - \Delta^i dS + r(\Delta S - V_1^i - \lambda V_2^i)dt \\
&= dV_1^i + dV_2^i - \Delta dS + r(\Delta S - V_1^i - \lambda V_2^i)dt \\
&= (\Theta_1^i + \lambda \Theta_2^i)dt + (\Delta_1^i + \lambda \Delta_2^i - \Delta^i)dS + \frac{1}{2}(\sigma S)^2(\Gamma_1^i + \lambda \Gamma_2^i)dt + r(\Delta S - V_1^i - \lambda V_2^i)dt \\
&= (\Theta_1^i + \lambda \Theta_2^i + r(\Delta S - V_1^i - \lambda V_2^i))dt \\
&= (\Theta_1^i - r(V_1^i - \Delta_1 S))dt + \lambda(\Theta_2^i - r(V_2^i - \Delta_2 S))dt \\
&= -\frac{1}{2}(\tilde{\sigma}_1 S)^2 \Gamma_1^i dt - \frac{\lambda}{2}(\tilde{\sigma}_2 S)^2 \Gamma_2^i dt \\
&= -\frac{1}{2}(\tilde{\sigma}_1 - \tilde{\sigma}_2)S^2 \Gamma_1^i
\end{aligned} \tag{131}$$

Integrating the present value of the mark-to-market profit of the delta-gamma neutral portfolio hedge from t_0 to T with respect to time gives

$$\Pi_T = \int_{t_0}^T d\Pi_{t_0}^i = \frac{1}{2}(\tilde{\sigma}_1^2 - \tilde{\sigma}_2^2) \int_{t_0}^T e^{-r(t-t_0)} S^2 \Gamma_1^i dt \tag{132}$$

Equation 132 is the total profit of the delta-gamma neutral strategy. Similar the total profit of the delta hedge this is estimated numerically using a discrete realisation of a log-normal random walk S_t , and because of the S_t term the profit is path dependent.

7.3 Theta

Theta measures the sensitivity in option value with respect to changes in time. Theta is essentially the erosion of the option premium through time. Theta for a European Call option is

$$\begin{aligned}
\Theta_C &= \frac{\partial C(S_t, t)}{\partial t} \\
&= \frac{\partial}{\partial t} (S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)) \\
&= S \frac{\partial}{\partial t} \Phi(d_1) - K \left(rKe^{-r\tau}\Phi(d_2) + Ke^{-r\tau} \frac{\partial}{\partial t} \Phi(d_2) \right) \\
&= -S\Phi'(d_1) \left(\frac{r + \frac{1}{2}\sigma^2}{\sigma\sqrt{\tau}} - \frac{1}{2\sqrt{\tau}}d_1 \right) - rKe^{-r\tau}\Phi(d_2) + Ke^{-r\tau} \left(\frac{r - \frac{1}{2}\sigma^2}{\sigma\sqrt{\tau}} - \frac{1}{2\sqrt{\tau}}d_2 \right)
\end{aligned}$$

Similar to the derivation of delta we note the identities $S\Phi'(d_1) = Ke^{-r\tau}\Phi'(d_2)$ and $d_2 = d_1 - \sigma\sqrt{\tau}$ to arrive at the theta for a call option.

$$\Theta_C = -S\Phi'(d_1) \frac{\sigma}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2) \tag{133}$$

Similarly, for a European put option we have

$$\Theta_P = -S\Phi'(-d_1)\frac{\sigma}{2\sqrt{\tau}} + rKe^{-r\tau}\Phi(-d_2) \quad (134)$$

Theta of an option contract under black-scholes contributes to the rate the option value decreases. For European call and put options, theta is largest in magnitude when the option is ATM, the reasoning is identical to that of gamma. This implies both Gamma and Theta are largest ATM.

7.4 Delta, Gamma and Theta

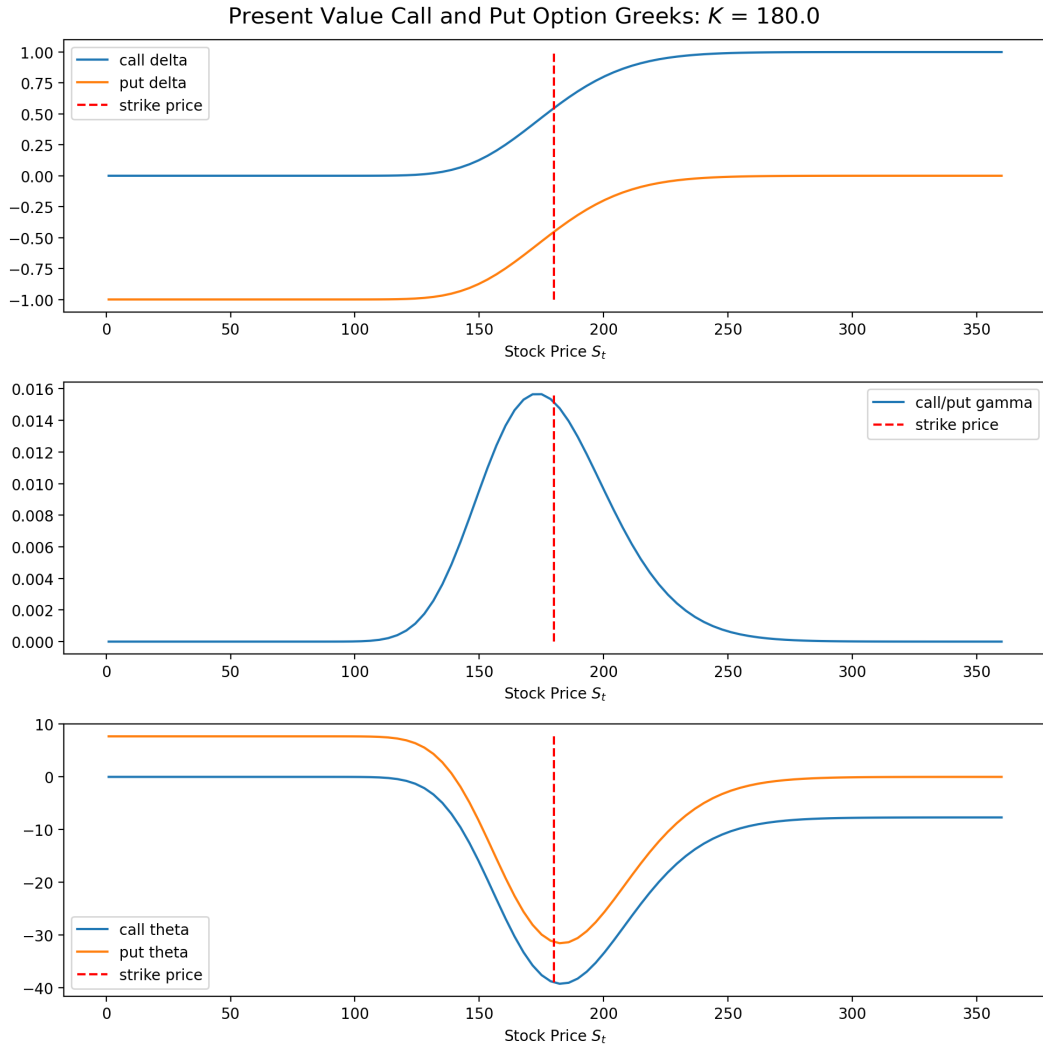


Figure 9: NVDA top of chain call and put contract 26-05-07. $K = 180, r = 0.04, \sigma = 0.39$

Figure 9 plots the three Greeks covered in this chapter, delta, gamma, and theta for the top of the chain call and put contracts for NVDA traded on 26-05-07. The delta for a put is effectively a shifted version of the delta for a call by -1. The delta is a CDF function and is therefore an increasing function of stock price with the most volatility around ATM.

The gamma for both calls and puts is identical, and is the density of the standard normal evaluated at d_1 . The gamma is maximised when the option is ATM, which makes sense given

gamma measures sensitivity of delta, this is largest when the option is ATM (this is when the rate of change of the delta is greatest). As t approaches T , the d_1 term explodes, and thus the CDF tends to a value of 1 if the option is INM or 0 if OTM. This is essentially a step function so the gamma tends to a Dirac delta, taking a max value at the exact point the option is ATM.

The theta for a call and put is negative, as the option value is a decreasing function of time. The theta for a call and put is most negative when the call or put is ATM, at this point d_1 takes its lowest value closest to 0, therefore maximising the density, the negative of which leads to the lowest theta value. If the call or put is deep OTM or ITM, due to symmetry, the density contributes less and the interest rate term dominates. If the option is deep ITM the CDF weights the discount term more, leading to a theta value slightly more negative than if the option were deep OTM, where the weight of the discount term would be approximately 0.

8 Black-Scholes Assumptions

Now that we have derived the Black-Scholes model for Vanilla call and put equity options, we can have a better understanding of the assumptions that underpin the model. The Black-Scholes model we presented in equation 32 makes the following set of assumptions, listed in no particular order

- **Log-normal Random Walk.** We assume the stock price follows a GBM. It follows that the solution follows a log-normal distribution defined by 24. We use this fact to solve for $df(S_t, t)$ using Ito's lemma.
- **No Arbitrage Pricing.** The market price of the option, V , must be equal to the portfolio that replicates the option's payoff throughout the option's life. This is the replicating portfolio from which we derive the Black-Scholes option price, $f(S_t, t)$. Otherwise, we can buy the overpriced option and sell the undervalued option and make an instantaneous profit equal to the mispricing. This is different from volatility arbitrage, where arbitrage opportunities arise from differences in implied and realised volatility.
- **Riskless Growth of Replicating Portfolio.** We assume the replicating portfolio grows deterministically according to the risk-free rate.
- **Continuous Time Delta Hedging.** We assume we can delta hedge continuously, so we can completely eliminate the stochastic term in our portfolio arising from dS_t .
- **No Dividends.** We assume the underlying pays no dividends as this would depreciate the stock price, S_t . Explicit solutions exist for equity options that pay dividends and the solutions for calls and puts are very similar to those defined in 91 and 95.

9 Conclusion

We have developed a fundamental understanding of the Black-Scholes option pricing formula from a mathematical and practical perspective and explored applications of the model other than just option pricing, such as delta hedging.

We began by building prerequisite knowledge essential to deriving the Black-Scholes PDE, specifically the properties of Brownian motion, Ito processes, and the generalised formula that allows us to model the evolution of such stochastic processes, Ito's lemma. We demonstrated the application of such techniques by deriving the solution to an Ito process that follows a GBM SDE.

Moving on to the focus of this paper, the Black-Scholes model, we began by deriving the Black-Scholes PDE by defining a replicating portfolio, and modelled its evolution as a deterministic process. We then eliminated the risk (realised volatility) in our portfolio by making a continuous delta-hedging argument to arrive at the Black-Scholes PDE. The following subchapters were dedicated to solving the Black-Scholes PDE for plain vanilla European non-dividend-paying call and put equity options. We began by applying a change of variables and an exponential transformation to the original PDE and then reversed time to transform the problem into the PDE for describing heat transfer in a medium, the heat equation. We then proceeded to solve the heat equation in the standard sense, applying Fourier transforms to represent the problem in the frequency domain, where it is a first-order linear system, solved using the integrating factor method. Undoing our transformation via an inverse Fourier transform yielded the general solution to our original Black-Scholes PDE.

Using this approach, we uncovered the solution for the option value was given by calculating the convolution between a Gaussian and the terminal value of the option. We then proved this yielded the famous call and put solution for the call and puts we were considering. An alternative equivalent method to solving the Black-Scholes PDE was also discussed, called the risk-neutral pricing method, which defines the current option value as a discounted expected future payoff of the contract under risk-neutral pricing.

Comparative statics for the Black-Scholes option solutions and their interpretations were also presented. We derive the delta, gamma and theta for both plain vanilla call and put options.

We finished by exploring volatility arbitrage and the profits that arise from employing delta hedging strategies using implied and realised volatility.

10 References

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